

Wang Xinyi

Address: 17W, Hong Kong Science Park, Shatin, N.T., Hong Kong.

E-mail: xywangmae@link.cuhk.edu.hk

Tel: (+852) 53921596

Education

- **Postdoctoral Fellow**, Department of Mechanical & Automation Engineering,
The Chinese University of Hong Kong 10/2023-Now
- **Ph.D. Degree**, Department of Mechanical & Automation Engineering,
The Chinese University of Hong Kong 08/2019-09/2023
Supervisor & Co-supervisor: Prof. Ben M. Chen (IEEE Fellow & Academy of Engineering Fellow,
Singapore) & Prof. Jie Huang (IFAC Fellow & IEEE Fellow)
- **B.Eng. Degree**, Flight Vehicle Design and Engineering, Xiamen University 09/2015-06/2019
GPA: 91.8/100 (Ranking 1st out of 46)
- **Visiting student**, University of Oxford 02/2017

Research interests

Task and motion planning, Multi-agent system, Trajectory optimization

Publications

(# = equal contribution, * = corresponding author)

● Journals

- [1] **X. Wang**, L. Xi, Y. Ding, B. M. Chen. Distributed Encirclement and Capture of Multiple Pursuers with Collision Avoidance. *IEEE Transactions on Industrial Electronics (TIE)*, 2023.
- [2] C. Gao, **X. Wang***, X. Chen, B. M. Chen. A Hierarchical Multi-UAV Cooperative Framework for Infrastructure Inspection and Reconstruction. *Control Theory and Technology (CTT)*, 2023.
- [3] **X. Wang**, L. Xi, Y. Chen, S. Lai, F. Lin, B. M. Chen. Decentralized MPC-based trajectory generation for multiple quadrotors in cluttered environments. *Guidance, Navigation and Control (GNC)*, 2021, 1(02): 2150007.
- [4] C. Gao, **X. Wang**, R. Wang, Z. Zhao, Y. Zhai, X. Chen, B. M. Chen. A UAV-based explore-then-exploit system for autonomous indoor facility inspection and scene reconstruction. *Automation in Construction (AIC)*, 2022.
- [5] L. Xi, **X. Wang**, L. Jiao, S. Lai, Z. Peng, B. M. Chen. GTO-MPC-Based Target Chasing Using a Quadrotor in Cluttered Environments. *IEEE Transactions on Industrial Electronics (TIE)*, 2021, 69(6): 6026-6035.

● Conference papers

- [1] **X. Wang**, J. Xu, C. Gao, Y. Chen, J. Zhang, C. Wang and B. M. Chen, Sensor-based multi-robot search and coverage with spatial separation in unstructured environments. To be presented at the 2024 IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, May 2024.
- [2] **X. Wang**, Y. Ding, Y. Chen, R. Han, L. Xi, B. M. Chen. OA-ECBVC: A Cooperative Collision-free Encirclement and Capture Approach in Cluttered Environments. *IEEE Conference on Decision and Control (CDC)*, 2023.
- [3] Y. Chen, R. Wang, **X. Wang***, B. M. Chen. Sampling-based path planning under temporal logic constraints with real-time adaptation. *IEEE International Conference on Robotics and Automation (ICRA)*, 2023.
- [4] **X. Wang***, C. Gao, X. Chen, B. M. Chen. Fast and Secure Distributed Multi-agent Coverage Control with An Application to Infrastructure Inspection and Reconstruction. *Chinese Control Conference (CCC)*, 2023, Guan Zhao-Zhi Award Finalist.
- [5] L. Xi#, **X. Wang#**, Ding Y, Y. Wei, Z. Peng, B. M. Chen. Quadrotor trajectory planning for visibility-aware

target following. IEEE International Conference on Robotics and Biomimetics (ROBIO), 2021.

[6] **X. Wang***, L. Xi, Y. Chen, S. Lai, F. Lin, B. M. Chen. Decentralized trajectory generation technique for multiple unmanned multi-copter systems in cluttered environments. The 12th International Micro Air Vehicle Conference (IMAV), 2021.

[7] Y. Chen, **X. Wang**, R. Wang, Z. Guo, S. Lai, B. M. Chen. An Interactive System for Multiple-task Linear Temporal Logic Path Planning. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023.

] R. Wang, Z. Guo, Y. Chen, **X. Wang** and B. M. Chen, Air bumper: A collision detection and reaction framework for autonomous MAV navigation. To be presented at the 2024 IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, May 2024.

● Patents

[1] B. M. Chen, Y. Chen, R. Wang, **X. Wang**, Xi Chen. A Human-in-the-Loop System for Mobile Robot Task Planning. Patent of the United States, No. 63/450,314, Mar. 6, 2023.

Research Grants

- **Collaborative Task Assignment of Multi-Agent Unmanned Systems for Infrastructure Inspection**, University Grants Committee, Hong Kong, 2024–2026. **Core Member**.
Develop a fast and secure distributed coverage control algorithm to enable a multi-agent system to cooperatively inspect infrastructure.
- **Collaborative Search and Pursuit-evasion for Unmanned Systems in Cluttered Environments**, University Grants Committee, Hong Kong, 2023-2025. **Core Member**.
Design an encirclement and capture strategy for multiple players and deploy it on physical ground robot systems to test performance.
- **Development of an Integrated Autonomous Building Inspection and Information Management System with Drones**, InnoHK Center for Logistics Robotics, Hong Kong, 2020–2025. **Core Member**.
Deploy 3 UAVs to inspect a large-scale industrial building in Hong Kong using our distributed coverage algorithm.
- **Key Technologies of UAV Systems for Field and Forest Operations**, Joint Fund of the National Natural Science Foundation of China, 2017–2020. **Project Participant**.
Design a guided time-optimal model predictive control based practical framework to generate chasing trajectories for the quadrotor and deploy it on Crazyflie platform to verify the performance.

Internships

Peng Cheng Laboratory & Shenzhen Weixi Technology Co., Ltd

05/2020-09/2020

Construct Navigation System of Four-wheel Skid-Steering Vehicle in a Park Environment with Pedestrians

- Construct MPC-based vehicle local planner with a trajectory tracking controller to solve problems in tracking the velocity command;
- Test real-time lane line path following and point to point delivery mission using lidar module with considering the dynamic obstacles and physical limitation of UGV.

Technical Skills

- Experienced in C/C++, Python, MATLAB, LaTeX, Git
- Robotics & Deep Learning: Robot Operating System (ROS), Gazebo, Pytorch

Courses & Teaching Experience

- IERG 5350 Reinforcement Learning, instructed by Professor Bolei Zhou.
- ENGG 5501 Foundations of Optimization, instructed by Professor Anthony Man-Cho So.
- ELEG 5491 Introduction to Deep Learning, instructed by Professor Hongsheng Li.

Teaching Assistant:

- MAEG 3080 Fundamentals of Machine Intelligence, instructed by Professor Juan Luis.
- ENGG 5402 - Advanced Robotics, instructed by Professor Fei Chen.

Awards and Honors

Postdoctoral Scholarship, the Innovation and Technology Fund of Hong Kong	10/2023-Now
Professor Charles K. Ko Student Creativity Award, CUHK	06/2023
Research Postgraduate Studentship, CUHK	08/2020
Outstanding Graduates, XMU (Top 2%)	06/2019
Merit Student, XMU (Twice, Top 8%)	1 2/2016&12/2017
Third Prize in RoboMaster 2017 (The 16th National College Students Robotics Contest)	08/2017
Xiamen International Bank Scholarship, XMU (Top 4%)	04/2017

Academic services

Reviewer of Journals

- IEEE Transactions on Automatic Control (TAC)
- IEEE Transactions on Cybernetics
- IEEE Robotics and Automation Letters (RA-L)
- Science China Information Science
- International Journal of Robust and Nonlinear Control
- Autonomous Intelligent Systems
- Unmanned Systems
- Guidance, Navigation and Control

Reviewer of Conferences

- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE Conference on Decision and Control (CDC)
- IEEE International Conference on Control and Automation (ICCA)
- IEEE International Conference on Unmanned Aircraft Systems (IEEE ICUAS 2022)
- Chinese Control Conference (CCC 2023)

Presentations

- ICRA 2023, London, England, poster presentation
- CDC 2023, Singapore, oral presentation
- CCC 2023, Tianjing, China, present for Guan Zhao-Zhi Award
- ICRA 2024, Yokohama, Japan, to be presented